

Edge-to-Cloud CPAP Video Coaching with Multimodal Deviation Detection

Pamit Duggal

DISP Laboratoire
Université Lumière Lyon 2,
Université Toulouse Capitole,
Lyon, France
pamitduggalfr@gmail.com

Yasaman Kakaei Siahkal

Univ Lumière Lyon 2, INSA Lyon,
Université Lyon 1, Université Jean
Monnet Saint-Étienne, DISP
Laboratoire, Linde HomeCare France,
Lyon, France
yasaman.kakaei-siahkal@univ-lyon2.fr

Nejib Moalla

Univ Lumière Lyon 2, INSA Lyon,
Université Lyon 1,
Université Jean Monnet Saint-Étienne,
DISP Laboratoire,
Lyon, France
Nejib.Moalla@univ-lyon2.fr

Olivier Grasset

Linde HomeCare France,
Bourg-en-Bresse, France
olivier.grasset@linde.com

Abstract—Continuous positive airway pressure (CPAP) telemonitoring reveals issues such as mask leaks, low nightly usage, and residual events in obstructive sleep apnea (OSA), but non-adherence and therapy dropout remain common. This paper describes a pipeline that goes beyond CPAP telemetry by adding digital biomarkers (blood-oxygen saturation, SpO₂, and heart-rate variability) and self-reported symptoms. These numbers are targets, not results already shown. We are aiming to cut dropout events by 20% and add 30 minutes of nightly use for patients at risk. Underneath, a quantized TensorFlow Lite (TFLite) classifier watches for abnormal or risky situations on a Raspberry Pi. Once it flags something, a second model, the deviation-to-video step, takes that issue, how severe it is, and what we know about the patient, and turns it into coaching content. Concretely, a Python script draws a chart straight from the clinical data. Then Google Veo animates it. To get the clip to the patient fast, delivery runs through a Quality-on-Demand (QoD) session on the CAMARA API, a Linux Foundation telecom project, aiming for 95% activation success and under 60 seconds of end-to-end latency when the asset is cached. Patients see the result on a bilingual dashboard, subtitled in English and French. We ran three test cases on real CPAP data and the pipeline held together end to end, though testing it on a larger group of real patients is still ahead of us.

Index Terms—CPAP adherence, OSA, edge AI, TensorFlow Lite, Quality-on-Demand, CAMARA API, generative video, VTT, telehealth, digital biomarkers, patient empowerment.

I. INTRODUCTION

CPAP is the standard treatment for moderate-to-severe OSA, but adherence to the therapy at home remains difficult to sustain [1], [2]. Telemonitoring platforms already give clinical teams data on usage, leaks, and residual events [2], [5], and programs that pair this data with a defined response pathway get better adherence than passive monitoring alone [4], [6]. The problem is that CPAP device data only tell part of the story: a leaking mask or a low-usage night shows up in the numbers, but nothing in that data explains why a patient is struggling, how they feel, or what kind of help would actually work. Adding patient-reported outcomes and digital biomarkers to the data layer is one way to move past simple monitoring [7].

Earlier internal work of ours linked CPAP telemetry to structured coaching responses, but left two problems open. This earlier approach had two limitations. First, the intervention logic relied on device metrics alone, so two patients landing in the same deviation class got identical responses, even when one had a completely different symptom history, fatigue level, or track record with past coaching. Second, when a situation did not improve after an intervention, there was no mechanism to determine why, which made it difficult to escalate care for patients who clearly needed more than the standard playbook. Three scenarios cause most of the trouble in practice: mask leaks that will not go away, chronically low nightly usage, and shifts that show up across multiple sensor streams at once. In all three, the root cause matters just as much as catching the deviation itself.

This paper presents a pipeline that connects patient-side sensing to personalized video coaching through two models running in sequence. The first, a multimodal deviation detection model, flags risky or abnormal situations using the combined data layer. The second, a Deviation-to-Video model, takes that flagged deviation, its severity, and the patient's context, and turns it into a coaching action. Four contributions come out of this work. First, a patient-side data layer that adds digital biomarkers, self-reported symptoms, and feedback from past coaching to CPAP telemetry, pulled from a set of connected-device backend APIs (Section III-B). Second, a multimodal deviation detection model built on TFLite and a Raspberry Pi that sorts sensor data into categories we can act on. Third, a Deviation-to-Video model where Python charts carry every clinical value and Google Veo does nothing but animate them. Fourth, a QoD-aware delivery system paired with a bilingual web dashboard.

II. STATE OF THE ART

Commercial CPAP apps such as ResMed myAir and Philips DreamMapper give patients a usage score and generic motivational messages. Telemonitoring call-centre services, of the kind described by Munafo [8], route alerts to a clinician

for a phone call. Neither approach personalizes content to a patient’s specific deviation, severity, or history, and neither uses physiological biomarkers or generated video. Recent generalist medical foundation models [11] open the door to LLM-driven coaching, but without a way to separate verified clinical values from generated text, they risk inventing medical information. Our pipeline handles this by keeping three things separate: an edge classifier over multimodal signals, a rule-and-retrieval layer that picks the coaching action, and a video stage where every clinical value is computed by deterministic code before the generative model ever sees it [11].

Work on 360-degree video and VR in healthcare finds that how useful a video feels depends on relevance, immersion, and clear guidance [9], and videos built from a patient’s own data, with concrete step-by-step explanations, tend to help self-management more than generic clips [9], [10].

On the technical side, quantized TFLite models can come close to full-precision accuracy, which makes a Raspberry Pi enough hardware for local analysis of physiological time series [12], and the CAMARA QoD API lets an application reserve bandwidth for a single event instead of holding a connection open continuously [13], [14]. Neither mechanism has been applied before to CPAP coaching specifically, which is the gap this paper addresses.

III. SYSTEM ARCHITECTURE

Four components carry patient-side data through to personalized video coaching, shown in Fig. 1. A Raspberry Pi edge node sits closest to the patient. It collects CPAP sensor data, digital biomarkers, and feedback, and it runs the TFLite deviation-detection model locally. From there, a cloud orchestration layer takes over, mapping confirmed events to coaching paths, managing the video database, and controlling the CAMARA QoD session. A generative video pipeline handles the actual content, producing a Python chart, a Google Veo animation, and bilingual VTT subtitle files. Finally, a bilingual patient dashboard plays the video back and logs whatever feedback the patient gives. We built each part behind its own interface, so swapping out a model, a media service, or a network feature does not mean touching anything else.

A. Patient-Side Data Layer

Usage duration and mask leak, on their own, catch the obvious treatment problems, but they provide no insight into what is happening for the patient at home overnight. A patient’s device readings can look completely normal while they are quietly getting more uncomfortable, more tired, or more discouraged by the week, and none of that shows up until adherence eventually drops. A high leak number by itself does not tell you whether the mask fits badly or the patient just sleeps on their stomach. That is why the data layer draws on four sources instead of one. Digital biomarkers such as SpO2 and heart-rate variability can catch early signs that therapy control is slipping. Self-reported symptoms from the dashboard, things like fatigue, headaches, a dry throat, or poor sleep quality, fill in what the sensors cannot see. Feedback on

Edge-to-cloud pipeline

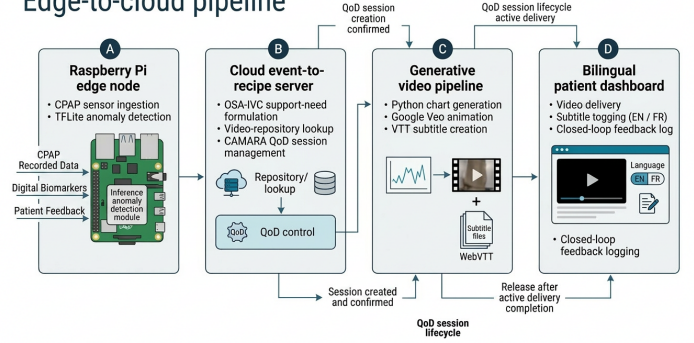


Fig. 1. Data-to-impact pipeline for CPAP video coaching.

TABLE I
BACKEND DATA SOURCES FOR THE PATIENT-SIDE LAYER

Endpoint	Source	Role
Usage	CPAP export	Airflow, leak, pressure
Intervention	EHR/coaching log	Prior interventions
Intervention definition	Config	Category/channel map
Monitoring	Dashboard	Questionnaires, ESS
Medical survey	Dashboard	Connected medical survey
Risk factor	Cohort DB	Risk-factor metrics
Biomarker enrollment	Cohort DB	Biomarker cohort IDs
Withings watch	Withings ScanWatch	SpO2, HRV, sleep
Withings BPM	Withings BPM Core	BP, ECG
Masimo	Masimo RadG	Oximetry, pulse
Hexoskin	Hexoskin ProShirt	Resp. effort, ECG
SomnoArt	SomnoArt	Sleep architecture, REM

past coaching tells us whether an earlier fix actually worked. Together, these let the system tell apart, for example, a patient who has leaks and reports discomfort from one who has leaks but feels fine about the device.

B. Device Compatibility and Data Sources

The data layer is fed through a defined set of backend API endpoints rather than one proprietary CPAP integration, so the pipeline stays device agnostic on the telemetry side while still pulling biomarkers from specific, validated wearables. Table I lists the endpoints used in the current build. CPAP airflow rate, oxygen-saturation trend, and mask-pressure variation come from standard telemonitoring exports (Usage), not a proprietary hardware protocol tied to a single manufacturer, so any CPAP platform with comparable export fields is expected to be compatible. Validating this compatibility against specific commercial CPAP brands remains future work. Biomarkers come from wearable-specific APIs with their own sampling schedules relative to the sleep session: continuous overnight recording for the Withings ScanWatch and the Hexoskin ProShirt, spot checks for the Withings BPM Core and the Masimo RadG. As for self-reported symptoms and post-coaching feedback, those come from dashboard forms that appear after each session, not from any kind of passive sensing.

TABLE II
TFLITE EDGE MODEL CONFIGURATION

Parameter	Value
Architecture	Multi-layer perceptron (MLP)
Input features	36 (12 per sensor channel)
Window size	20 s at 1 Hz
Output classes	3 (Normal / Alarming / Digital biomarker)
Quantisation	INT8 post-training
Model size (TFLite)	<500 kB
Training data	Real CPAP data, anomaly injection
Target hardware	Raspberry Pi 5 (ARM Cortex-A76)

C. Edge Node: Multimodal Deviation Detection

The edge node runs on a Raspberry Pi 5 and classifies the patient’s current situation using the multimodal deviation detection model and the combined data layer. Nothing raw ever leaves this device, which keeps both bandwidth use and privacy exposure in check. Only the aggregated event summary makes its way to the cloud. Three CPAP sensor channels are collected: airflow rate, oxygen-saturation trend, and mask-pressure variation, split into 20-second windows sampled at 1 Hz. For each window, the feature extractor computes mean, variance, maximum excursion, and zero-crossing rate per channel, giving a 36-dimensional input vector for a quantized TFLite multi-layer perceptron (Table II) that labels the window Normal, Alarming, or Digital-biomarker. A sustained mask leak or a sharp SpO2 drop triggers an alarm. Deviations linked across several channels over a sustained period get flagged as a digital-biomarker event. When the classifier’s confidence in a non-normal class crosses the operating threshold, the edge node packages the event and sends it to the cloud over a secured REST connection.

D. Cloud Orchestration: Deviation-to-Video Model

The cloud orchestration layer turns a confirmed deviation, its severity, and the patient’s context into a video recipe, working as a rule-and-retrieval system rather than a learned model. Building a video can go one of three ways. If a validated asset already matches the event type and the patient’s context, the system just retrieves it. If something close exists but is not quite right, it adapts that asset by updating the chart data, threshold notes, and subtitles. And if nothing suitable is on file, it kicks off a full generation workflow and saves whatever comes out for next time. Where the event goes depends on what triggered it: a mask leak sends the patient down a mask-fit coaching path, low usage routes to a motivation-focused path built around their usage trends and recent feedback, and a digital-biomarker event escalates straight to a technician for review.

E. Programmatically Anchored Video Pipeline

The video pipeline implements the Deviation-to-Video model by keeping clinical facts separate from the animation [11]. A Python script pulls the sensor-window data and clinician-defined thresholds for the confirmed event and draws

a full-HD 16:9 dark-mode Matplotlib chart with sensor traces, threshold lines, and the anomalous segments marked. Every value, colour, and label in that chart comes from deterministic code. That static chart goes to the Google Veo Vertex API as an image-to-video prompt that only specifies camera movement, pacing, and style, producing a 15-to-30-second animated clip. A separate script writes VTT subtitle files in English and French, timed to the video, and both the clip and the subtitles are saved to the content repository for reuse.

F. Bilingual Patient Dashboard

Patients access their coaching video through a web application, which displays English and French subtitles as soon as the video becomes available. They can switch between the two languages or replay the clip without breaking playback. Once they have watched it, a short prompt asks whether they think the problem is fixed, and their answer decides what happens next: the case closes as resolved, it repeats, or it gets escalated to a technician. We hold onto all of that feedback to retrain both models down the line.

IV. IMPLEMENTATION AND EVALUATION

A. Model Training and Data

We train the deviation-detection model on real CPAP data streams, injecting anomaly patterns for each of the three deviation classes [5], [7]. High leak and brief oxygen drops make up the alarming cases. Digital-biomarker cases are messier: deviations show up across several channels at once, which usually points to something getting worse rather than a one-off glitch. We train the model at full precision first. Then we convert it to INT8 TFLite for the edge deployment. The Deviation-to-Video model, by contrast, is not trained at all. It is a rule set built around the three coaching paths and thresholds our clinician partners approved [12].

B. End-to-End Scenarios

We tested the full pipeline against three cases: mask leak, low use, and digital biomarker. A test harness sends predefined classification sequences to the edge component to trigger each deviation class, and every case checks both the QoD grant and its fallback. Table III shows what actually completed at each stage.

All stages ran as expected in the three cases, and visual checks confirmed that no clinical values leaked into the Veo layer. Table IV reports the system-level KPIs measured on the Raspberry Pi 5 edge node across the three scenarios. Preliminary live-deployment logs on the Raspberry Pi 5 also separate edge processing time from phone-to-edge network latency. An early observation suggests the two vary independently, which a future latency-distribution study should characterize separately. The targets came from practical usability limits rather than statistical inference: 60 seconds is a reasonable wait for cached playback on a patient dashboard, and the 2% stall ratio follows common streaming-quality guidance. With only three sessions, the "3/3" figures show that the pipeline works, not that it is statistically reliable.

TABLE III
PIPELINE COMPLETION PER SUPPORT CASE (✓ = PASS, – = NOT APPLICABLE)

Stage	Leak	Low use	Digital biomarker
Edge detection	✓	✓	✓
Event-to-recipe mapping	✓	✓	✓
Repository lookup	✓(retrieve)	✓(adapt)	–
Vevo generation	–	–	✓
VTT EN + FR	✓	✓	✓
QoD granted	✓	✓	✓
QoD fallback	✓	✓	✓
Web app delivery	✓	✓	✓
Feedback / escalation	✓(closed)	✓(closed)	✓(esc.)

TABLE IV
TECHNICAL KPI RESULTS ON RASPBERRY PI 5 ACROSS SCENARIOS (N=3 PILOT SESSIONS, A FEASIBILITY CHECK, NOT A STATISTICALLY POWERED ESTIMATE)

KPI	Target	Observed
End-to-end delay (cached asset)	≤60 s	<60 s ✓
End-to-end delay (generated clip)	≤15 min	<15 min ✓
QoD activation success rate	≥95%	3/3 sessions ✓
Delivery reliability	≥95%	3/3 cases ✓
Playback stall ratio	≤2%	0% ✓
Edge resource saving (vs cloud)	≥20%	cache path ✓

V. DISCUSSION

The principal contribution of this work is the integrated chain itself, rather than any single component within it. Edge AI, generative video, and QoD-aware delivery are stitched together around a data layer that goes past CPAP metrics into digital biomarkers, self-reported symptoms, and coaching feedback. Individually, none of these components is novel. A small MLP is not novel, and neither is a rule-and-retrieval mapper, Google Vevo, or the CAMARA QoD API. The contribution lies in how these components are integrated, rather than in a new learning algorithm. Keeping clinical content and generated animation strictly apart protects against the video inventing medical information it should not, and only reserving QoD bandwidth for confirmed events ensures that network resources are not reserved unnecessarily, while coaching is still delivered promptly when clinically relevant.

VI. CONCLUSION

This paper described a pipeline for personalized CPAP video coaching that handles three clinical scenarios: mask leak, low nightly use, and digital-biomarker events. What we have built is a full chain, from patient-side sensing through multimodal deviation detection and Deviation-to-Video translation, all the way to QoD-aware delivery and a bilingual dashboard. It ran correctly across all three scripted test cases on real CPAP data, so we have a working software prototype in hand. What remains absent is evidence that the system detects deviations accurately in a clinical sense or that it affects adherence. No sensitivity, specificity, ROC/AUC, confusion matrix, or

comparison against a threshold-rule baseline is reported. This needs to happen before we can say anything about detection quality in a clinical sense. The 20% dropout-reduction and 30-minute usage-increase figures are targets for a future prospective study, not results from this one. A structured focus group with patients and clinicians, together with a prospective study on real device and biomarker data, are the next steps, along with a proper false-alarm and failure-mode analysis of the classifier.

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REFERENCES

- [1] A. J. Watach, D. Hwang, and A. M. Sawyer, “Personalized and patient-centered strategies to improve PAP adherence in OSA patients,” *Patient Prefer. Adherence*, vol. 15, pp. 1557–1570, 2021.
- [2] B. K. S. Thong et al., “Telehealth technology in enhancing CPAP adherence in OSA,” *Front. Med.*, vol. 9, p. 877765, 2022.
- [3] J. Lacroix et al., “Effectiveness of digitally generated personalized feedback on CPAP adherence: RCT,” *J. Med. Internet Res.*, vol. 25, p. e40193, 2023.
- [4] S. R. Patel et al., “Impact of an extended telemonitoring and coaching program on CPAP adherence,” *Ann. Am. Thorac. Soc.*, vol. 19, no. 12, pp. 2070–2076, 2022.
- [5] L. Calé-Ségura et al., “Management of CPAP follow-up by telemonitoring in OSA: the PROTEUS project,” *J. Clin. Med.*, 2025.
- [6] P. Schisano et al., “Distance follow-up by a remote medical care centre improves CPAP adherence,” *Sleep Med.*, 2024.
- [7] H. Lin et al., “Impact of a multimodal telemonitoring intervention on CPAP adherence,” *Chest*, 2021.
- [8] R. Munafo, “How telemonitoring improves timeliness of care and CPAP adherence,” *Sleep Review*, 2020.
- [9] C. Laker, P. Knight-Davidson, and A. McVicar, “The use of 360-degree video to reduce anxiety in mental health nursing students,” *Nurs. Rep.*, vol. 15, no. 5, p. 157, 2025.
- [10] C. Lauritzen et al., “Development of therapeutic competencies in health care students: qualitative focus group study using 360-degree video and virtual reality technology,” *JMIR Med. Educ.*, vol. 12, e75776, 2026.
- [11] M. Moor et al., “Foundation models for generalist medical AI,” *Nature*, vol. 616, pp. 259–265, 2023.
- [12] R. David et al., “TensorFlow Lite Micro: embedded ML for TinyML systems,” *Proc. MLSys*, vol. 3, pp. 800–811, 2021, M. Zanghieri et al., “Robust real-time embedded EMG recognition with TCNs,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 14, no. 2, pp. 244–256, 2020.
- [13] CAMARA Project, “Quality-on-Demand (QoD) API specification,” Linux Foundation, 2024.
- [14] R. Lumb et al., “5G for healthcare: enabling new care models,” *IEEE Netw.*, vol. 34, no. 2, pp. 48–54, 2020.